

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 July 2026
Team ID	Team 1
Project Name	Employee Software Installation Request through Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The architectural diagram below shows the components and information flow of the solution.

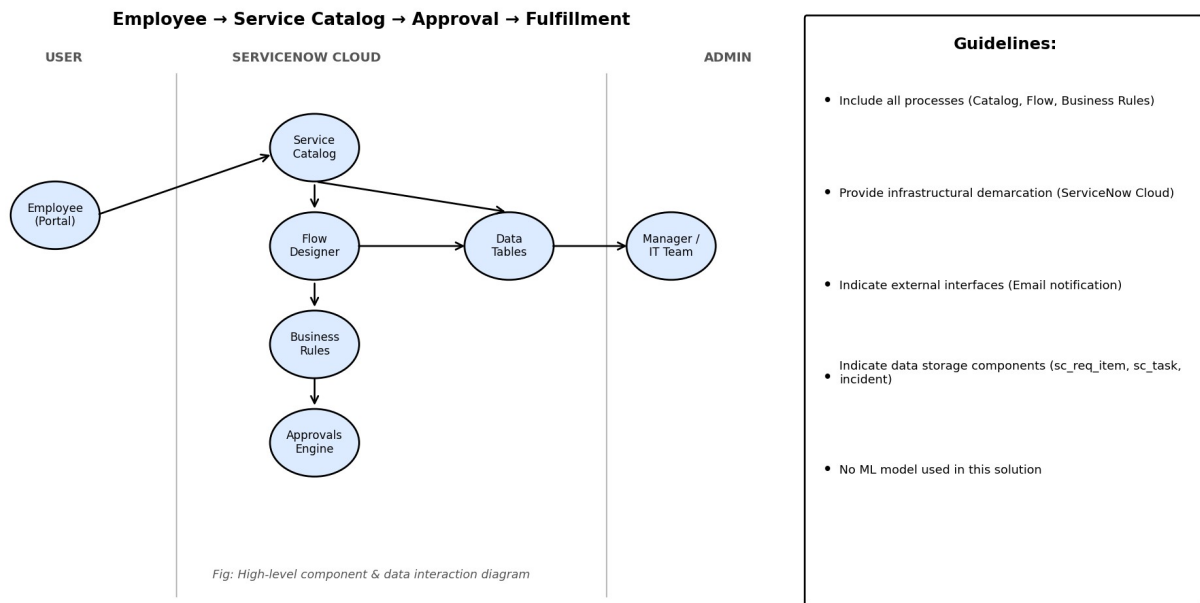


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the employee interacts with the application	ServiceNow Service Portal / Now Experience UI
2	Application Logic-1	Approval routing and branching logic	Flow Designer
3	Application Logic-2	Automated Incident creation on license shortage	Business Rules (GlideRecord / JavaScript)
4	Application Logic-3	Role-based access restrictions	Access Control Lists (ACL)
5	Database	Core platform data storage	ServiceNow tables (sc_req_item, sc_task, incident, sys_user)
6	Cloud Database	Hosted instance data	ServiceNow Cloud (Now Platform)
7	File Storage	Attachment storage (if applicable)	ServiceNow Attachment table (sys_attachment)
8	External API	Not used in current scope	N/A
9	Machine Learning Model	Not used in current scope	N/A
10	Infrastructure	Application deployment	ServiceNow Personal Developer Instance (Cloud-hosted)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Not applicable — built on the proprietary ServiceNow platform	N/A
2	Security Implementations	Role-based ACLs, Cross-scope Access control, single-approver validation	ServiceNow ACL engine, Cross-scope Access

3	Scalable Architecture	Native multi-tenant cloud platform scales without infrastructure changes	ServiceNow Cloud
4	Availability	Managed hosting with platform-level uptime SLA	ServiceNow Cloud hosting
5	Performance	Asynchronous Flow execution avoids blocking user actions	Flow Designer engine

References:

<https://docs.servicenow.com>

<https://developer.servicenow.com>

<https://www.servicenow.com/products/service-catalog.html>